**UNIVERSITY ALUMNI MANAGEMENT SYSTEM**

**Development, Design and System Architecture Report**

**Prepared By**

**Developer:** Student Project Team

**Technologies Used**

- Vite.js
- Tailwind CSS
- JavaScript
- HTML5
- CSS3

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**1. INTRODUCTION**

The University Alumni Management System is a web-based platform developed to improve communication, networking, and engagement between alumni, students, and the university administration. The system provides a centralized digital environment where graduates can create professional profiles, share job opportunities, participate in university events, and mentor current students.

Traditional alumni management approaches often rely on manual records and fragmented communication channels, which limit collaboration and networking opportunities. This project addresses these challenges by introducing an interactive and scalable web solution developed using modern frontend technologies such as Vite.js and Tailwind CSS.

The system is designed to enhance alumni participation, simplify information management, and strengthen relationships between graduates and the university community.

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**2. PROBLEM STATEMENT**

Many universities face difficulties in maintaining communication with alumni after graduation. Existing systems are often outdated, decentralized, or entirely manual, making it difficult to track alumni activities, provide mentorship opportunities, and organize networking events.

Students also lack accessible platforms for connecting with experienced alumni who can guide them academically and professionally.

Therefore, there is a need for a modern web-based system that can:

- Maintain updated alumni records
- Facilitate networking and mentorship
- Share employment opportunities
- Improve university-alumni engagement

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**3. OBJECTIVES**

**3.1 Main Objective**

To develop a web-based University Alumni Management System that improves communication, networking, and engagement between alumni, students, and the university.

**3.2 Specific Objectives**

- To enable alumni registration and profile management
- To allow sharing of job opportunities and events
- To provide mentorship connections between alumni and students
- To create a centralized alumni database
- To improve communication between the university and graduates

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## 4. SCOPE OF THE SYSTEM

The system focuses on:

- Alumni registration and login
- Alumni profile management
- Event announcements
- Job posting and applications
- Student-alumni mentorship
- Admin management dashboard

The system does not include:

- Financial transactions
- Academic grading systems
- Student admission processing

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## 5. SYSTEM REQUIREMENTS

### 5.1 Functional Requirements

The system shall:

- Allow users to register and log in
- Allow alumni to update their profiles
- Enable administrators to manage users
- Allow posting and viewing of job opportunities
- Enable event creation and participation
- Allow students to search for alumni mentors

### 5.2 Non-Functional Requirements

The system shall:

- Be user-friendly
- Be responsive on mobile and desktop devices
- Support scalability
- Maintain secure authentication
- Provide fast page loading

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## 6. DEVELOPMENT METHODOLOGY

The project follows the Agile Software Development Methodology.

**Agile Phases**

**Requirement Gathering**

Information about alumni challenges and communication gaps was collected.

**System Planning**

Project structure, technologies, and architecture were defined.

**Design**

Wireframes, UI layouts, and database structures were created.

**Development**

Frontend components were developed using:

- Vite.js
- Tailwind CSS
- JavaScript

**Testing**

The system was tested for:

- Functionality
- Responsiveness
- Usability
- Performance

**Deployment**

The system is prepared for deployment on cloud hosting platforms.

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## 7. SYSTEM ARCHITECTURE

### 7.1 Architecture Overview

The system follows a Client-Server Architecture.

**Components**

**Frontend Layer**

Developed using:

- Vite.js
- Tailwind CSS
- JavaScript

Responsibilities:

- User interaction
- Data presentation
- Form validation
- Responsive UI rendering

**Backend Layer**

Handles:

- Authentication
- API processing
- Business logic

Possible technologies:

- Node.js + Express
- Flask API
- Firebase

**Database Layer**

Stores:

- User profiles
- Alumni records
- Events
- Job posts
- Messages

Possible databases:

- MySQL
- MongoDB
- Firebase Firestore

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**7.2 System Architecture Diagram**

```
+-------------------+
|   Web Browser    |
+-------------------+
        |
```

```
        v
+-------------------+
|   Frontend (Vite) |
| Tailwind + JS UI  |
+-------------------+
        |
        v
+-------------------+
|    Backend API    |
| Authentication    |
| Business Logic    |
+-------------------+
        |
        v
+-------------------+
|     Database      |
| Alumni Records    |
| Jobs & Events     |
+-------------------+
```

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**8. SYSTEM DESIGN**

**8.1 Use Case Design**

**Actors**

- Alumni
- Students
- Administrator

**Main Use Cases**

- Register account
- Login
- Update profile
- Post jobs

- View events
- Request mentorship
- Manage users

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### 8.2 Navigation Structure

Home

├── About

├── Alumni Directory

├── Jobs

├── Events

├── Mentorship

├── Login

└── Register

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## 9. DATABASE DESIGN

### Main Tables

### Users Table

| Field | Type |
|---|---|
| id | Integer |
| fullname | Varchar |
| email | Varchar |
| password | Varchar |
| role | Varchar |

### Alumni Profiles Table

| Field | Type |
|---|---|
| profile_id | Integer |
| user_id | Integer |
| graduation_year | Integer |
| profession | Varchar |

| Field | Type |
| --- | --- |
| company | Varchar |

**Jobs Table**

| Field | Type |
| --- | --- |
| job_id | Integer |
| title | Varchar |
| description | Text |
| posted_by | Integer |

**Events Table**

| Field | Type |
| --- | --- |
| event_id | Integer |
| title | Varchar |
| location | Varchar |
| event_date | Date |

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## 10. USER INTERFACE DESIGN

### Design Principles

The interface was designed with:

- Simplicity
- Accessibility
- Responsiveness
- Modern aesthetics

### Tailwind CSS Features Used

- Utility-first styling
- Responsive breakpoints
- Flexbox and Grid layouts
- Reusable UI components

### Main Pages

#### Landing Page

Displays:

- Introduction
- Quick navigation
- Featured alumni

**Dashboard**

Displays:

- User profile
- Notifications
- Job opportunities
- Upcoming events

**Alumni Directory**

Allows searching and filtering alumni profiles.

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## 11. FUNCTIONAL MODULES

### 11.1 Authentication Module

Handles:

- Registration
- Login
- Password encryption
- Session management

### 11.2 Alumni Management Module

Handles:

- Alumni profiles
- Career information
- Professional networking

### 11.3 Job Portal Module

Handles:

- Job posting
- Job browsing
- Applications

### 11.4 Event Management Module

Handles:

- Event announcements
- RSVPs
- Event tracking

**11.5 Mentorship Module**

Handles:

- Student requests
- Mentor matching
- Communication

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**12. DEVELOPMENT ENVIRONMENT**

**Software Tools**

| Tool | Purpose |
|---|---|
| VS Code | Code editor |
| Vite.js | Frontend build tool |
| Tailwind CSS | Styling framework |
| GitHub | Version control |
| Node.js | Runtime environment |

**Installation Commands**

**Create Vite Project**

npm create vite@latest alumni-system

**Install Dependencies**

cd alumni-system

npm install

**Install Tailwind CSS**

npm install -D tailwindcss postcss autoprefixer

npx tailwindcss init -p

**Run Development Server**

npm run dev

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## 13. SECURITY CONSIDERATIONS

The system implements:

- Password hashing
- Input validation
- Secure authentication
- Role-based access control
- Session protection

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## 14. TESTING AND VALIDATION

**Testing Methods**

**Unit Testing**

Individual components tested separately.

**Integration Testing**

Modules tested together.

**User Acceptance Testing**

Real users tested the system usability.

**Expected Results**

- Fast page loading
- Responsive layouts
- Secure login
- Accurate data management

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## 15. DEPLOYMENT STRATEGY

The system can be deployed using:

- Vercel
- Netlify
- Firebase Hosting
- Render

**Deployment Steps**

1. Build project

```
npm run build
```

2. Upload generated files to hosting server
3. Configure domain and SSL certificate

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**16. FUTURE IMPROVEMENTS**

Future enhancements may include:

- Mobile application support
- AI-based mentorship recommendations
- Video conferencing integration
- Real-time chat system
- Alumni donation system
- Analytics dashboard

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**17. CONCLUSION**

The University Alumni Management System provides an efficient and modern solution for strengthening alumni relationships and improving communication between graduates, students, and the university administration.

Using Vite.js and Tailwind CSS ensures a fast, scalable, and responsive frontend application capable of supporting future growth and additional features. The system improves alumni engagement, supports mentorship opportunities, and creates a sustainable networking platform for the university community.

The project demonstrates the practical application of modern web technologies in solving communication and management challenges within educational institutions.